

Vulnerability Assessment Report

Vulnerability Assessment Report using Nmap and OWASP ZAP

Introduction

This report presents a vulnerability assessment performed using Nmap and OWASP ZAP to analyze the network and web application security of a target system. The purpose of this assessment is to identify exposed services, potential security risks, and provide recommendations for improvement.

Tools Used

- Nmap (Network scanning and port analysis tool)
 - OWASP ZAP (Web application vulnerability scanner)
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Target

<http://scanme.nmap.org>

Nmap Scan Results

The following open ports were identified during the scan:

- 22/tcp (SSH - Secure Shell)
- 80/tcp (HTTP - Web service)
- 9929/tcp (nping-echo service)
- 31337/tcp (Elite service)

Some ports were found to be filtered, indicating that firewall rules are in place to restrict access.

OWASP ZAP Scan Results

OWASP ZAP performed an automated scan on the target website. The scan identified informational and low-risk security issues. No critical or high-risk vulnerabilities were detected, as the target is a controlled testing environment designed for learning purposes.

Risk Analysis

Risk Level	Description
Low	Minor informational issues
Medium	Issues that require attention
High	Critical vulnerabilities requiring immediate action

Overall risk level: Low to Medium

Recommendations

- Disable unnecessary or unused services
 - Strengthen SSH authentication (use key-based authentication instead of passwords)
 - Implement security headers such as HSTS and CSP
 - Regularly monitor and scan open ports and services
 - Use firewalls to restrict unauthorized access
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Conclusion

The vulnerability assessment using Nmap and OWASP ZAP provided insights into the security posture of the target system. While some services were exposed, no critical vulnerabilities were identified. The system is suitable for testing purposes but would require additional hardening in a production environment.